Course Code	Course Title	L	T	P	C
BMHA314E	Autonomous Mobile Robots	3	0	2	4
Pre-requisite	BMHA303L, BMHA303P	Syllabus version			
		2.0			

Course Objectives

1. To understand the fundamental principles underlying autonomous mobile robots, including perception, localization, mapping, path planning, and control.
2. To develop practical skills in implementing algorithms for autonomous navigation and control using simulation environments and real robotic platforms.
3. To identify challenges and limitations in autonomous navigation and develop problem-solving skills to address complex issues in autonomous mobile robotics.

Course Outcomes

On the successful completion of the course the students will be able to

1. Understand the fundamentals and various types of autonomous mobile robots.
2. Formulate Kinematics and dynamic model of mobile robots.
3. Select appropriate sensors and perception algorithms for autonomy
4. Develop localization and mapping algorithms that enable autonomous mobile robots to accurately determine their position and build maps of their surroundings.
5. Design and execute autonomous navigation tasks, including path planning, obstacle avoidance, and dynamic environment interaction.
6. Understand the fundamentals of multi-robot systems.

Module:1	Introduction	5 hours

Introduction to mobile robots and mobile manipulators - Principle of locomotion and types of locomotion - Types of mobile robots: ground robots (wheeled and legged robots), aerial robots, underwater robots, and water surface robots.

Module:2	Kinematics and Dynamics	8 hours

Kinematics of wheeled mobile robot - degree of freedom and maneuverability - generalized wheel model - different wheel configurations - holonomic and non-holonomic robots - Forward and inverse kinematics of mobile robots - Dynamics modeling and analysis -Trajectory planning and control.

Module:3	Perception for Autonomous Navigation	7 hours

Sensors for Perception - Lidar (Light Detection and Ranging), Radar (Radio Detection and Ranging), IMU (Inertial Measurement Unit), Wheel Encoders – Perception Algorithms - Visual Tracking & Servoing - Sensor fusion techniques - Object detection and recognition - Environmental understanding and scene interpretation.

Module:4	Localization and Mapping	7 hours

Techniques for localization: odometry, landmark-based, probabilistic methods - Map representations: grid-based, feature-based, topological maps - Localization

algorithms: Kalman filters, Particle filters - Map building and maintenance - Simultaneous Localization and Mapping (SLAM)		
Module:5	**Path Planning and Navigation**	**6 hours**
Path planning algorithms: Dijkstra, A*, RRT, potential fields - Metaheuristic and non-heuristic algorithms - Nonholonomic Motion Planning - Collision free path planning - Local and global path planning - Dynamic obstacle avoidance - Navigation control architectures - Integration with perception and localization systems.		
Module:6	**Control and Decision Making**	**6 hours**
Control strategies for mobile robots: PID, model predictive, reinforcement learning - Behavior-based control architectures - Decision making under uncertainty - Task planning and scheduling - Safety and reliability considerations		
Module:7	**Multi robot systems and collaboration**	**4 hours**
Multi-robot systems and coordination - Swarm robots, cooperative and collaborative robots - Human-robot interaction - Autonomous exploration - Learning-based approaches for autonomy		
Module:8	**Contemporary Issues**	**2 hours**
	Total Lecture hours:	**45 hours**

Text Book(s)

1.	R Siegwart, IR Nourbakhsh, D Scaramuzza, Introduction to Autonomous Mobile Robots, MIT Press, USA, 2011.
2.	Kelly, A, Mobile Robotics: Mathematics, Models, and Methods, Cambridge University Press, 2013.

Reference Books

1.	SG Tzafestas, Introduction to Mobile Robot Control, Elsevier, USA, 2014.
2.	Hughes, C. and Hughes, T., Robot programming: a guide to controlling autonomous robots. Que Publishing, 2016

Mode of Evaluation: CAT, Assignment, Projects, Quiz, FAT

List of Experiments (Indicative)

1.	Setting up the environment in ROS and understanding the fundamentals.
2.	Simulate a mobile robot model (e.g., TurtleBot) in Gazebo
3.	Implement SLAM (Simultaneous Localization and Mapping) using the ROS Navigation Stack
4.	Design and implement a path planning algorithm using ROS navigation packages (e.g., move_base).
5.	Implement object detection and recognition using ROS perception libraries (e.g., OpenCV, PCL).
6.	Set up communication between multiple robots using ROS topics, services, or actionlib.